

x930 Series

Advanced Gigabit Layer 3 Stackable Switches with 10G and 40G Uplinks



The Allied Telesis x930 Series of stackable Gigabit Layer 3 switches provide resiliency, reliability and high performance, making them ideal for distribution and network core solutions.

x930 Series switches are a high-performing and feature-rich choice for today's networks. With a choice of 24- and 48-port models with 10 Gigabit and 40 Gigabit uplink ports, plus the power of Allied Telesis Virtual Chassis Stacking (VCStack™) with up to 160Gbps of stacking bandwidth per switch, the x930 Series have the flexibility and performance for key network connectivity.

Network automation

Vista Manager™ EX bundled with Allied Telesis Autonomous Management Framework™ Plus (AMF Plus) meets the increasing management requirements of modern networks. While AMF Plus allows an entire network to be securely and easily managed as a single virtual device, Vista Manager EX provides an intuitive and powerful graphical tool for monitoring and managing AMF Plus wired and Autonomous Wave Control (AWC) wireless devices.

Device, network, and security management

The Device GUI on the x930 Series enables graphical monitoring of key switch features to support easy management.

Integrated into the Device GUI, the wireless controller supports visibility and management of AMF Plus wired and AWC wireless network devices, making it ideal as a one-stop solution for small to medium-sized networks.

AWC is an intelligent, easy to use Wireless LAN controller that automatically maintains optimal wireless coverage. The wireless controller includes AWC floor and heat maps showing wireless coverage. It also supports AWC Channel Blanket hybrid operation, providing maximum performance and seamless roaming, as well as AWC Smart Connect for simplified deployment, and a resilient Wi-Fi network solution using wireless uplink connectivity.

Resilient

Allied Telesis Ethernet Protection Switched Ring (EPSRing™), G.8032 Ethernet Ring Protection, and Media Redundancy Protocol (MRP) ensure that distributed ring-based network segments have resilient access to online resources.

Allied Telesis Virtual Chassis Stacking (VCStack™), in conjunction with link aggregation, provides a network with no single point of failure for highavailability applications. The x930 Series can stack up to eight¹ units for enhanced resiliency and simple device management. Plus, Long Distance Stacking (VCStack LD) allows stacks to be created over fiber links, making the x930 the perfect choice for distributed environments too.

Reliable

The x930 Series was designed with reliability in mind, and guarantees continual delivery of essential services. With dual hot-swappable load-sharing power supplies and near-hitless online stack reconfiguration, maintenance may be performed without affecting network uptime.

Secure

Advanced security features protect the network from the edge to the core. The x930 Series offers powerful control over network traffic types, protection against network attacks, secure management options, loop guard to detect cabling mistakes, and tri-authentication for comprehensive end-point access control.

Future proof

The x930 Series ensures a futureproof network, with superior flexibility and ability to stack multiple units, plus 10 Gigabit and 40 Gigabit uplink ports. The x930 Series is Software Defined Networking (SDN) ready, supporting OpenFlow



Key Features

- Allied Telesis Autonomous Management Framework™ Plus (AMF Plus)
- AMF-Security compatible
- Bidirectional Forwarding Detection (BFD)
- VCStack™ up to 8¹ switches
- VCStack LD for long distance stacking
- EPSR Master
- G.8032 Ethernet Ring Protection
- Continuous PoE
- Precision Time Protocol (PTP) Transparent Mode
- 40G Ethernet uplink/stacking ports
- Active Fiber Monitoring (AFM) for fiber data and stacking links
- OpenFlow for SDN
- Upstream Forwarding Only (UFO)
- VLAN Translation
- Media Access Control Security (MACSec)
- Media Redundancy Protocol (MRP)
- Modbus support
- Multicast Source Discovery Protocol (MSDP)
- Link Monitoring
- The wireless controller enables:
 - Wired and wireless network visibility
 - AWC wireless network management
 - AWC-Channel Blanket hybrid wireless
 - AWC-Smart Connect wireless uplinks
- NETCONF/RESTCONF with YANG data modelling

v1.3 and a comprehensive IPv6 feature set to ensure they are ready for future network traffic demands.

¹ Up to 4 units supported if using 1Gbps ports for stacking

KEY FEATURES

Wireless controller

Integrated into the Device GUI, the wireless controller provides full network visibility of AMF Plus wired and AWC wireless devices. Manage and simplify wireless deployment with AWC-Smart Connect, and support optimal wireless performance from AWC hybrid operation with maximum throughput and a seamless Wi-Fi user experience.

Allied Telesis Autonomous Management Framework™ Plus (AMF Plus)

AMF Plus is a sophisticated suite of management tools that provide a simplified approach to network management. Powerful features like centralized management, auto-backup, auto-upgrade, auto-provisioning and auto-recovery enable plug-and-play networking and zero-touch management.

Any x930 Series switch can operate as the AMF Plus network master, storing firmware and configuration backups for other network nodes. The AMF Plus master enables auto-provisioning and auto-upgrade by providing appropriate files to new network members. New network devices can be pre-provisioned making installation easy because no on-site configuration is required.

AMF Plus secure mode encrypts all AMF traffic, provides unit and user authorization, and monitors network access to greatly enhance network security.

An AMF Plus license (from AW+ version 5.5.2-2 onwards) provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards)

AWC Wireless Management

Optimize wireless network performance with the Autonomous Wave Controller (AWC), built-in to the x930 Series. AWC analyzes wireless traffic patterns and automatically reconfigures access points to meet demand.

Wireless network operation in multi-channel, single-channel (Channel Blanket), and hybrid (multi-channel and Channel Blanket) modes, supports maximum data throughput and seamless roaming for the most flexible wireless solution available.

AWC-Smart Connect (AWC-SC) enables plug-and play wireless network growth, as new APs only need a power connection, and will then automatically create resilient wireless uplink connections to other APs.

Virtual Chassis Stacking (VCStack™)

Create a VCStack of up to eight¹ units with 40Gbps (or 160Gbps with the StackQS model) of stacking bandwidth on each unit. Stacking links are connected in a ring so each device has dual connections to further improve resiliency. VCStack provides a highly available system where network resources are spread out across stacked units, reducing the impact if one of the units fails. Aggregating switch ports on different units across the stack provides excellent network resiliency.

Long Distance Stacking (VCStack LD)

Long distance stacking allows a VCStack to be created over longer distances, perfect for a distributed network environment.

Bidirectional Forwarding Detection (BFD)

BFD enables fast detection of link failures, so recovery time is minimized. BFD works with static routes, and also alongside BGP and OSPF dynamic routing protocols supporting faster shutdown of neighbor connections if a peer session goes down. When using VRF-Lite, BFD is supported globally or within a domain.

Ethernet Protection Switched Ring (EPSRing™)

EPSRing and 10 Gigabit Ethernet allow several switches to form high-speed protected rings capable of recovery within as little as 50ms. The x930 Series switches can act as the EPSR Master.

This feature is perfect for high performance and high availability at the core of enterprise or provider access networks.

Superloop Protection enables a link between two EPSR nodes to be in separate EPSR domains, improving redundancy and network fault resiliency.

G.8032 Ethernet Ring Protection

G.8032 provides standards-based high-speed ring protection, that can be deployed stand-alone, or interoperate with Allied Telesis EPSR.

Ethernet Connectivity Fault Monitoring (CFM) proactively monitors links and VLANs, and provides alerts when a fault is detected.

NETCONF/RESTCONF

NETCONF/RESTCONF with YANG data modeling provides a standardized way to represent data and securely configure devices.

Media Recovery Protocol (MRP)

MRP enables high-availability automation networks, and is specified for rings with up to 50 devices, where it guarantees fully deterministic switchover behavior.

Virtual Routing and Forwarding (VRF-Lite)

VRF-Lite provides Layer 3 network virtualization by dividing a single switch into multiple independent virtual routing domains. With independent routing domains, IP addresses can overlap without causing conflict, allowing multiple customers to have their own secure virtual network within the same physical infrastructure. VRF-Lite supports IPv4 and IPv6 unicast and multicast traffic.

The built-in DHCP Server on the x930 Series is VRF aware, enabling the supply of IP addresses to clients across multiple isolated networks.

Active Fiber Monitoring

Active Fiber Monitoring prevents eavesdropping on fiber communications by monitoring received optical power. If an intrusion is detected, the link can be automatically shut down, or an operator alert can be sent. Active Fiber Monitoring is supported on fiber data and fiber stacking links.

UniDirectional link Detection

UniDirectional Link Detection (UDLD) is useful for monitoring fiber-optic links between two switches that use two single-direction fibers to transmit and receive packets. UDLD prevents traffic from being sent across a bad link by blocking the ports at both ends of the link in the event that either the individual transmitter or receiver for that connection fails.

Power over Ethernet Plus (PoE+)

With PoE, a separate power connection to media endpoints such as IP phones and wireless access points is not necessary. PoE+ reduces costs and provides even greater flexibility, providing the capability to connect devices requiring more power (up to 30 Watts) such as pan, tilt and zoom security cameras.

Continuous PoE

Continuous PoE allows the switch to be restarted without affecting the supply of power to connected devices. Smart lighting, security cameras, and other PoE devices will continue to operate during a software upgrade on the switch.

Modbus

Modbus enables communication with Supervisory Control and Data Acquisition (SCADA) systems for industrial automation.

High Reliability

The x930 series switches feature front to back cooling and dual power supply units (PSUs). The x930 features dual hot-swappable load sharing power supplies for maximum uptime, and the option of either front-to-back or back-to-front cooling. This makes it ideal for use as a top-of-rack data center switch.

VLAN Mirroring (RSPAN)

VLAN mirroring allows traffic from a port on a remote switch to be analysed locally. Traffic being transmitted or received on the port is duplicated and sent across the network on a special VLAN.

sFlow

sFlow is an industry-standard technology for monitoring high-speed switched networks. It provides complete visibility into network use, enabling performance optimization, usage accounting/billing, and defense against security threats. Sampled packets sent to a collector (up to 5 collectors can be configured) ensure it always has a real-time view of network traffic.

Precision Time Protocol (PTP)

PTP (IEEE 1588v2) synchronizes clocks throughout the network with micro-second accuracy, supporting industrial automation and control systems.

Software Defined Networking (SDN)

OpenFlow is a key technology that enables the use of SDN to build smart applications that unlock value and reduce cost.

AMF-Security and Application Proxy

The AMF-Security (AMF-Sec) solution enables internal LAN threat detection and automatic endpoint isolation to protect the network. The AMF Application Proxy enables the AMF-Sec controller to communicate with the AMF Plus master when a threat is detected, so the AMF Plus master can take action to block the threat at source by quarantining the infected endpoint.

Upstream Forwarding Only (UFO)

UFO lets you manage which ports in a VLAN can communicate with each other, and which only have upstream access to services, for secure multi-user deployment.

¹ Up to 4 units supported if using 1Gbps ports for stacking

VLAN Translation

VLAN Translation allows traffic arriving on a VLAN to be mapped to a different VLAN on the outgoing paired interface.

In Metro networks, it is common for a network Service Provider (SP) to give each customer their own unique VLAN, yet at the customer location give all customers the same VLAN-ID for tagged packets to use on the wire. SPs can use VLAN Translation to change the tagged packet's VLAN-ID at the customer location to the VLAN-ID for tagged packets to use within the SP's network.

This feature is also useful in Enterprise environments where it can be used to merge two networks together, without manually reconfiguring the VLAN numbering scheme. This situation can occur if two companies have merged and the same VLAN-ID is used for two different purposes.

Media Access Control Security (MACSec)

802.1AE MACSec secures all traffic on point-to-point Ethernet links between directly connected nodes, ensuring protection against security threats such as denial of service, intrusion, man-in-the-middle, passive wiretapping, and playback attacks.

Multicast Source Discovery Protocol (MSDP)

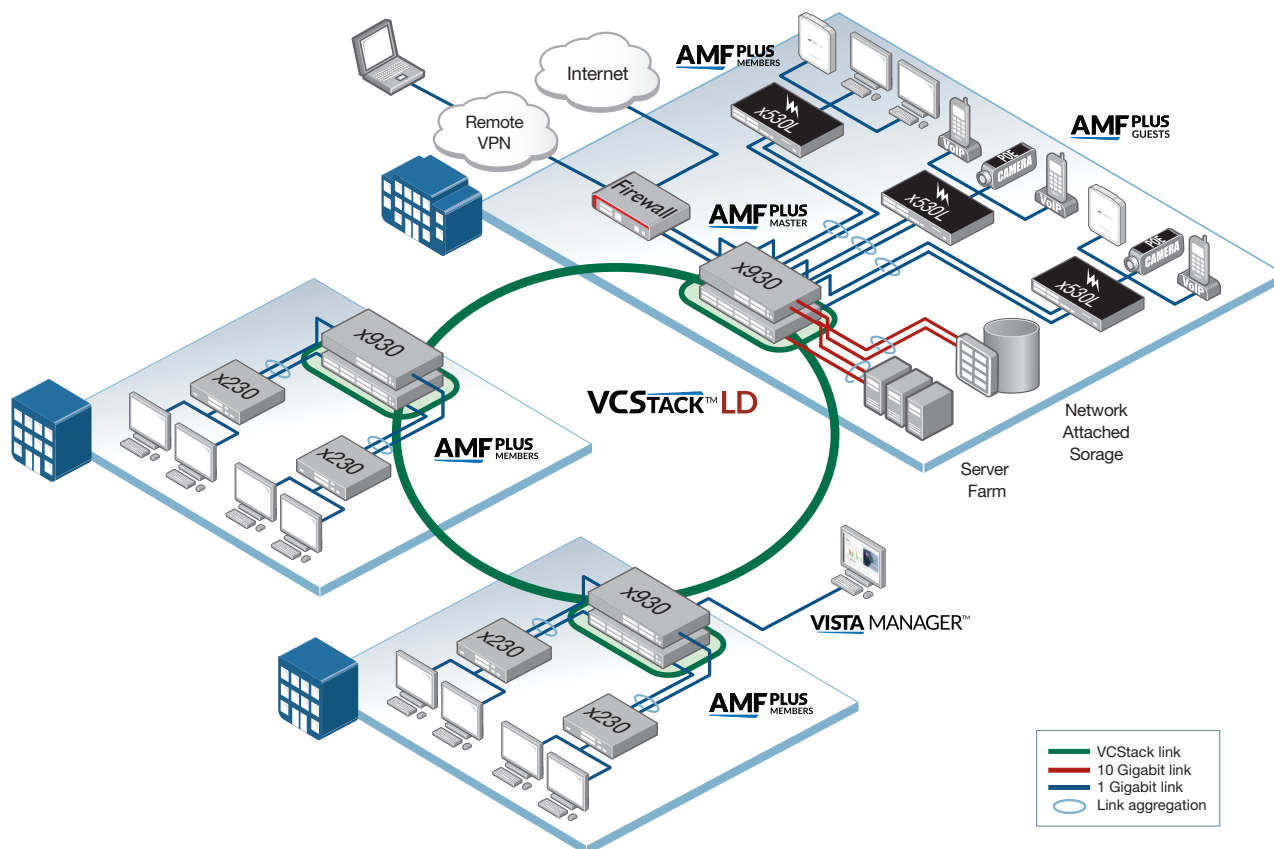
MSDP enables two or more PIM-SM (Sparse Mode) domains to share information on active multicast sources, for more efficient forwarding of multicast traffic.

Link Monitoring (Linkmon)

Linkmon enables network health monitoring by regularly sending probes over key links to gather metrics comprising latency, jitter, and probe loss. This supports pro-active network management, and can also be used with triggers to automate a change to device or network configuration in response to the declining health of a monitored link.



Distributed network core



Allied Telesis x930 Series switches are ideal for core and distributed solutions, where resiliency and flexibility are required. In the above diagram, long distance Virtual Chassis Stacking (VCStack LD) is used to create a single virtual unit out of multiple devices. The increased distance provided by fiber stacking connectivity means that members of the virtual chassis do not need to be co-located. Instead, they

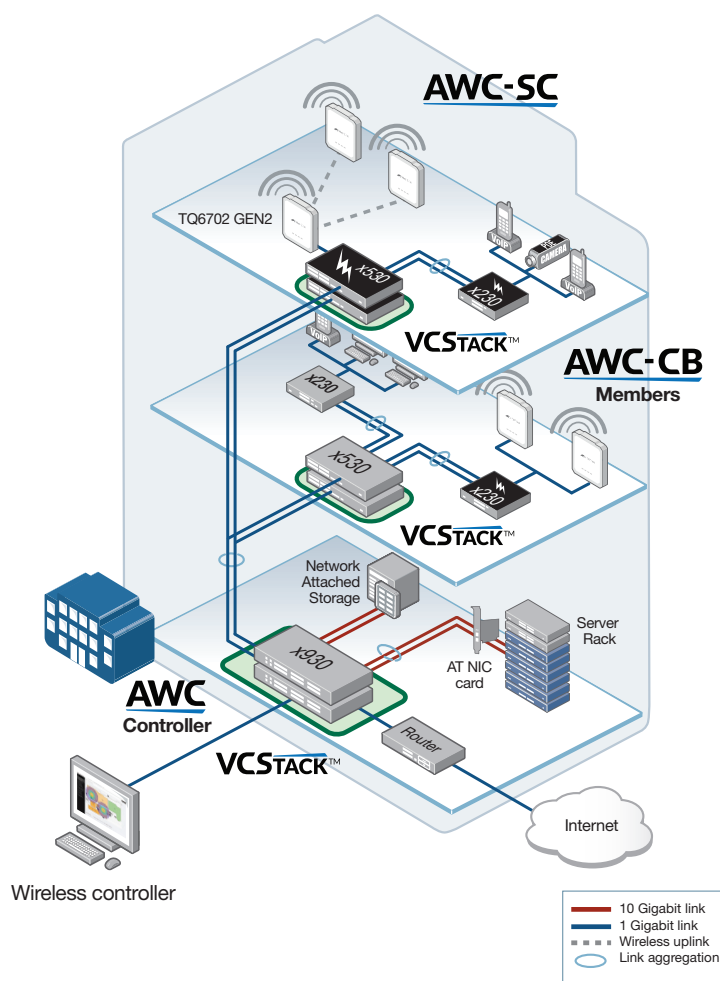
can be kilometers apart—perfect for a distributed network environment.

When combined with link aggregation to access switches, this provides a solution with no single point of failure that fully utilizes all network bandwidth, and ensures high availability of data for network users.

AMF Plus allows this large distributed network to be managed as a single virtual entity, greatly reducing administration and automating many day-to-day tasks.

Allied Telesis x930 Series switches support enterprises and their use of business-critical online resources and applications, with a resilient and reliable solution.

Integrated wireless LAN management



Allied Telesis Autonomous Wave Controller (AWC) offers solutions for two of the most common problems with Wireless LANs: initial setup complexity and on-going performance degradation. Initial WLAN set-up usually requires a site survey to achieve the best coverage; and performance of WLANs can often change over time as external sources of radio interference reduce coverage and bandwidth. These issues can be time-consuming to identify and resolve.

AWC features an intelligent process that automatically re-calibrates the signal strength and radio channel of each Access Point (AP) for optimal WLAN performance.

AWC Smart Connect (AWC-SC) uses wireless uplink connections between APs, so deployment is as easy as plugging in and powering on the new APs, which automatically extend the Wi-Fi network, creating a resilient solution.

The wireless controller is integrated into the Device Gui of the x930 Series and provides an ideal solution for modern enterprise networks, enabling management of both the wired (with AMF Plus) and wireless (with AWC) networks to be automated. This reduces both the time and cost of network administration, as well as maximizing network performance for a superior user experience.

Up to 5 TQ Series wireless APs can be managed for free, and up to a further 120 APs (max 125) with feature licenses, available separately.

On some AP models, hybrid channel blanket enables multi-channel and single-channel WiFi operation simultaneously. This supports seamless roaming and maximum throughput. Channel Blanket licenses are available for up to 120 APs. For plug-and-play wireless deployment AWC-SC licenses are available for up to 120 APs.

SPECIFICATIONS

	10/100/1000T (RJ-45) copper ports	100/1000X SFP ports	1/10 gigabit SFP+ ports	10 Gigabit stacking ports	Module slots	PoE enabled ports	Switching Fabric	Forwarding Rate
x930-28GTX	24	-	4 (2 if stacked)	2*	1	-	288Gbps	214.3Mpps
x930-28GPX	24	-	4 (2 if stacked)	2*	1	24	288Gbps	214.3Mpps
x930-28GSTX	24 (combo)	24 (combo)	4 (2 if stacked)	2*	1	-	288Gbps	214.3Mpps
x930-52GTX	48	-	4 (2 if stacked)	2*	1	-	336Gbps	250Mpps
x930-52GPX	48	-	4 (2 if stacked)	2*	1	48	336Gbps	250Mpps

* Stacking ports can be configured as additional 1G/10G Ethernet ports when unit is not stacked, or if StackQS module is used

Performance

- 40Gbps of stacking bandwidth per switch using front panel 10G SFP+ ports
- 160Gbps of stacking bandwidth per switch using optional AT-StackQS expansion module
- 13KB L2 and 9KB L3 Jumbo frames
- Wirespeed multicasting
- 4094 configurable VLANs
- Up to 64K MAC addresses
- Up to 16,000 OSPF routes
- Up to 2K IPv4 multicast entries
- Up to 2000 OpenFlow v1.3 entries
- Up to 128 Link Aggregation Groups (LAGS) - any combination of static and dynamic (LACP)
- 2GB DDR SDRAM, 256MB flash memory
- Packet buffer memory: x930-28 - 2MB, 52 - 4MB

Reliability

- Modular AlliedWare Plus operating system
- Internal dual hot-swappable PSUs, providing uninterrupted power and extra reliability
- Full environmental monitoring of PSUs, fans, temperature and internal voltages. SNMP traps alert network managers in case of any failure
- MTBF (Mean Time Between Failures):
 - x930-28GTX: 330,000 hours
 - x930-28GPX: 300,000 hours
 - x930-28GSTX: 280,000 hours
 - x930-52GTX: 280,000 hours
 - x930-52GPX: 240,000 hours
 - x9EM/XT4: 2,380,000 hours
 - FAN09: 850,000 hours
 - PWR150: 140,000 hours
 - PWR250: 210,000 hours
 - PWR800: 560,000 hours
 - PWR1200: 230,000 hours

Expandability

- Stack up to eight¹ units in a VCStack
- Versatile licensing options for additional features

Flexibility and Compatibility

- Gigabit SFP ports on x930-28GSTX will support any combination of Allied Telesis 100Mbps SFP and 10GbE SFP+ modules listed in this document under Ordering Information
- 10G SFP+ ports will support any combination of Allied Telesis 1000Mbps SFP and 10GbE SFP+ modules and direct attach cables listed in this document under Ordering Information
- Port speed and duplex (full duplex only) configuration can be set manually or by auto-negotiation
- Front-panel SFP+ stacking ports can be configured as additional 1G/10G Ethernet ports

Diagnostic Tools

- Active Fiber Monitoring detects tampering on optical links
- Built-In Self Test (BIST)
- Cable fault locator (TDR)
- Connectivity Fault Management (CFM) Continuity Check Protocol (CCP) for use with G.8032 ERPS
- Find-me device locator
- Hardware health monitoring
- Automatic link flap detection and port shutdown
- Optical Digital Diagnostic Monitoring (DDM)
- Ping polling and TraceRoute for IPv4 and IPv6
- Port mirroring
 - No limit on mirrored ports
 - Up to 4 mirror (analyzer) ports
- VLAN mirroring (RSPAN)
- Uni-Directional Link Detection (UDLD)

IPv4 Features

- Black hole routing
- Directed broadcast forwarding
- DNS relay
- Equal Cost Multi Path (ECMP) routing
- Policy-based routing
- Route maps and route redistribution (OSPF, BGP, RIP)
- Static unicast and multicast routing for IPv4
- UDP broadcast helper (IP helper)
- Up to 64 Virtual Routing and Forwarding (VRF-Lite) domains (with license)

IPv6 Features

- DHCPv6 client and relay
- DNS client and relay
- IPv4 and IPv6 dual stack
- IPv6 over IPv4 tunneling (manual configuration only)
- IPv6 aware storm protection, QoS and hardware ACLs
- Device management over IPv6 networks with SNMP, Telnet and SSH
- NTP client and server
- Static unicast and multicast routing for IPv6
- Log to IPv6 hosts with Syslog
- IPv6 Ready certified
- VRF-Lite

Management

- Front panel 7-segment LED provides at-a-glance status and fault information
- Allied Telesis Management Framework Plus (AMF Plus) enables powerful centralized management and zero-touch device installation and recovery
- Try AMF Plus for free with the built-in Starter license (includes network management and automation features, but not Vista Manager AMF Plus features)

- NETCONF/RESTCONF northbound interface with YANG data modelling
- Console management port on the front panel for ease of access
- Eco-friendly mode allows ports and LEDs to be disabled to save power
- Web-based Graphical User Interface (GUI)
- Industry-standard CLI with context-sensitive help
- Out-of-band 10/100/1000T Ethernet management port
- Comprehensive SNMP MIB support for standards-based device management
- Built-in text editor and powerful CLI scripting engine
- Event-based triggers allow user-defined scripts to be executed upon selected system events
- USB interface allows software release files, configurations and other files to be stored for backup and distribution to other devices

Quality of Service

- 8 priority queues with a hierarchy of high priority queues for real time traffic, and mixed scheduling, for each switch port
- Limit bandwidth per port or per traffic class down to 64kbps
- Wirespeed traffic classification with low latency essential for VoIP and real-time streaming media applications
- IPv6 QoS support
- Policy-based QoS based on VLAN, port, MAC and general packet classifiers
- Policy-based storm protection
- Extensive remarking capabilities
- Taildrop for queue congestion control
- Queue scheduling options for strict priority, weighted round robin or mixed scheduling
- IP precedence and DiffServ marking based on layer 2, 3 and 4 headers

Resiliency Features

- BPDU forwarding
- 10G and 40G stacking ports can be configured as Ethernet ports
- Control Plane Prioritization (CPP) ensures the CPU always has sufficient bandwidth to process network control traffic
- EPSRing (Ethernet Protection Switched Rings) with SuperLoop Protection (SLP) and enhanced recovery for extra resiliency
- G.8032 Ethernet Ring Protection
- Media Redundancy Protocol (MRP)
- Bidirectional Forwarding Detection (BFD)
- Long-Distance VCStack (LD-VCStack) using SFP+ or QSFP+ modules
- Loop protection: loop detection and thrash limiting
- PVST+ compatibility mode
- STP root guard
- VCStack fast failover minimizes network disruption

¹ Up to 4 units supported if using 1Gbps ports for stacking

Security Features

- Access Control Lists (ACLs) based on layer 3 and 4 headers
- Configurable ACLs for management traffic
- Dynamic ACLs assigned via port authentication
- ACL Groups enable multiple hosts/ports to be included in a single ACL, reducing configuration
- Auth fail and guest VLANs
- Authentication, Authorisation and Accounting (AAA)
- Bootloader can be password protected for device security
- BPDU protection
- DHCP snooping, IP source guard and Dynamic ARP Inspection (DAI)
- DoS attack blocking and virus throttling
- Dynamic VLAN assignment
- MAC address filtering and MAC address lock-down
- Media Access Control Security (MACSec)
- Network Access and Control (NAC) features manage endpoint security

- Learn limits (intrusion detection) for single ports or LAGs
- Private VLANs provide security and port isolation for multiple customers using the same VLAN
- RADIUS group selection per VLAN or port
- Simple Certificate Enrollment Protocol (SCEP) supports secure management
- Secure Copy (SCP) and Secure File Transfer Protocol (SFTP) client
- Strong password security and encryption
- TACACS+ command authorisation
- Tri-authentication: MAC-based, web-based and IEEE 802.1x

Software Defined Networking (SDN)

- OpenFlow v1.3 with support for encryption, connection interruption and inactivity probe

Environmental Specifications

- Operating temperature range:
0°C to 50°C (32°F to 122°F) AT-x930-GTX models and AT-x930-28GSTX

- 0°C to 45°C (32°F to 113°F) AT-x930-GPX models Derated by 1°C per 305 meters (1,000 ft)
- Storage temperature range:
-25°C to 70°C (-13°F to 158°F)
- Operating relative humidity range:
5% to 90% non-condensing
- Storage relative humidity range:
5% to 95% non-condensing
- Operating altitude:
3,048 meters maximum (10,000 ft)

Electrical Approvals and Compliances

- EMC: EN55022 class A, FCC class A, VCCI class A, ICES-003 class A
- Immunity: EN55024, EN61000-3-levels 2 (Harmonics), and 3 (Flicker) – AC models only

Power Supply Requirements

- AC voltage: 90 to 260V (auto-ranging)
- Frequency: 47 to 63Hz
- DC voltage: 40 to 60VDC (for PWR250-80 PSU only)

Physical Specifications

	Width	Depth	Height	Mounting	Unpackaged Weight	Packaged Weight
x930-28GTX	440 mm (17.32 in)	420 mm (16.54 in)	44 mm (1.73 in)	Rack-mount	5.1 kg (11.2 lb)	7.1 kg (15.7 lb)
x930-28GPX	440 mm (17.32 in)	420 mm (16.54 in)	44 mm (1.73 in)	Rack-mount	5.1 kg (11.2 lb)	7.1 kg (15.7 lb)
x930-28GSTX	440 mm (17.32 in)	420 mm (16.54 in)	44 mm (1.73 in)	Rack-mount	5.1 kg (11.2 lb)	7.1 kg (15.7 lb)
x930-52GTX	440 mm (17.32 in)	420 mm (16.54 in)	44 mm (1.73 in)	Rack-mount	5.1 kg (11.2 lb)	7.1 kg (15.7 lb)
x930-52GPX	440 mm (17.32 in)	420 mm (16.54 in)	44 mm (1.73 in)	Rack-mount	5.2 kg (11.5 lb)	7.2 kg (15.9 lb)
StackQS	141 mm (5.56 in)	96.5 mm (3.80 in)	40.3 mm (1.59 in)	Module	0.2 kg (0.44 lb)	1.2 kg (2.65 lb)
x9EM/XT4	141 mm (5.56 in)	96.5 mm (3.80 in)	40.3 mm (1.59 in)	Module	0.2 kg (0.44 lb)	1.2 kg (2.65 lb)

Power and Noise Characteristics

	No POE Load			Full POE+ Load (PWR800)			Full POE+ Load (PWR1200)		
	Max Power Consumption	Max Heat Dissipation	Noise	Max Power Consumption	Max Heat Dissipation	Noise	Max Power Consumption	Max Heat Dissipation	Noise
x930-28GTX	84W	285 BTU/h	39.7 dBA	-	-	-	-	-	-
x930-28GPX	84W	286 BTU/h	44.7 dBA	56.4W	287 BTU/h	45.8 dBA	808W	301 BTU/h	56.0 dBA
x930-28GSTX	97W	329 BTU/h	39.7 dBA	-	-	-	-	-	-
x930-52GTX	95W	323 BTU/h	39.7 dBA	-	-	-	-	-	-
x930-52GPX	97W	330 BTU/h	44.7 dBA	577W	331 BTU/h	45.8 dBA	880W	341 BTU/h	56.0 dBA

Noise: tested to ISO7779; front bystander position

Power over Ethernet Power supply Combinations

	POE Power Available	Maximum POE Ports Supported				Max Redundant POE Power
		Class 1 (4.0W)	Class 2 (7.0W)	Class 3 (15.4.0W)	Class 4 (30W)	
PWR800	380W	48	48	24	12	
PWR800 v2	380W	48	48	24	12	
PWR800 + PWR800	740W	48	48	48	24	380W
PWR800 v2 + PWR800 v2	740W	48	48	48	24	380W
PWR1200	740W	48	48	48	24	
PWR1200 v2	740W	48	48	48	24	
PWR1200 (v1 or v2) + PWR1200 (v1 or v2)	1440W	48	48	48	48	740W

Note: The x930 does not support mixed use of PWR800 and PWR800 v2

Latency (microseconds)

	Port Speed				
	10Mbps	100Mbps	1Gbps	10Gbps	40Gbps
x930-28GTX/GPX	47.4µs	47.4µs	3.7µs	2.6µs	-
x930-28GSTX	47.4µs	7.6µs (Fiber)	3.6µs (Fiber)	2.6µs	-
x930-52GTX/GPX	47.4µs	7.9µs	3.7µs	2.6µs	-
StackQS	-	-	-	-	2.5µs
x9EM/XT4	-	-	3.7µs	2.6µs	-

Safety

- Standards: UL60950-1, CAN/CSA-C22.2 No. 60950-1-03, EN60950-1, EN60825-1, AS/NZS 60950.1
- Certification: UL, cUL, TUV

Restrictions on Hazardous Substances (RoHS) Compliance

- EU RoHS compliant
- China RoHS compliant

STANDARDS & PROTOCOLS

AlliedWare Plus Operating System

Version 5.5.6-1

Border Gateway Protocol (BGP)

- BGP dynamic capability
BGP outbound route filtering
- RFC 1772 Application of the Border Gateway Protocol (BGP) in the Internet
- RFC 1997 BGP communities attribute
- RFC 2385 Protection of BGP sessions via the TCP MD5 signature option
- RFC 2439 BGP route flap damping
- RFC 2545 Use of BGP-4 multiprotocol extensions for IPv6 inter-domain routing
- RFC 2858 Multiprotocol extensions for BGP-4
- RFC 2918 Route refresh capability for BGP-4
- RFC 3392 Capabilities advertisement with BGP-4
- RFC 3882 Configuring BGP to block Denial-of-Service (DoS) attacks
- RFC 4271 Border Gateway Protocol 4 (BGP-4)
- RFC 4360 BGP extended communities
- RFC 4456 BGP route reflection - an alternative to full mesh iBGP
- RFC 4724 BGP graceful restart
- RFC 4893 BGP support for four-octet AS number space
- RFC 5065 Autonomous system confederations for BGP

Cryptographic Algorithms

FIPS Approved Algorithms

Encryption (Block Ciphers):

- AES (ECB, CBC, CFB and OFB Modes)
3DES (ECB, CBC, CFB and OFB Modes)

Block Cipher Modes:

- CCM
CMAC
GCM
XTS

Digital Signatures & Asymmetric Key Generation:

- DSA
ECDSA
RSA

Secure Hashing:

- SHA-1
SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512)
Message Authentication:
HMAC (SHA-1, SHA-2(224, 256, 384, 512)
Random Number Generation:
DRBG (Hash, HMAC and Counter)

Non FIPS Approved Algorithms

- RNG (AES128/192/256)
DES
MD5

Ethernet

- IEEE 802.1AE Media Access Control Security (MACSec)
- IEEE 802.2 Logical Link Control (LLC)
- IEEE 802.3 Ethernet
- IEEE 802.3ab 1000BASE-T
- IEEE 802.3ae 10 Gigabit Ethernet
- IEEE 802.3af Power over Ethernet (PoE)
- IEEE 802.3an 10GBASE-T
- IEEE 802.3az Energy Efficient Ethernet (EEE)
- IEEE 802.3ba 40GBASE-X
- IEEE 802.3u 100BASE-X
- IEEE 802.3x Flow control - full-duplex operation
- IEEE 802.3z 1000BASE-X

IEEE 1588v2 Precision clock synchronization protocol v2

IPv4 Features

- RFC 768 User Datagram Protocol (UDP)
- RFC 791 Internet Protocol (IP)
- RFC 792 Internet Control Message Protocol (ICMP)
- RFC 793 Transmission Control Protocol (TCP)
- RFC 826 Address Resolution Protocol (ARP)
- RFC 894 Standard for the transmission of IP datagrams over Ethernet networks
- RFC 919 Broadcasting Internet datagrams
- RFC 922 Broadcasting Internet datagrams in the presence of subnets
- RFC 932 Subnetwork addressing scheme
- RFC 950 Internet standard subnetting procedure
- RFC 951 Bootstrap Protocol (BootP)
- RFC 1027 Proxy ARP
- RFC 1035 DNS client
- RFC 1042 Standard for the transmission of IP datagrams over IEEE 802 networks
- RFC 1071 Computing the Internet checksum
- RFC 1122 Internet host requirements
- RFC 1191 Path MTU discovery
- RFC 1256 ICMP router discovery messages
- RFC 1518 An architecture for IP address allocation with CIDR
- RFC 1519 Classless Inter-Domain Routing (CIDR)
- RFC 1542 Clarifications and extensions for BootP
- RFC 1591 Domain Name System (DNS)
- RFC 1812 Requirements for IPv4 routers
- RFC 1918 IP addressing
- RFC 2581 TCP congestion control

IPv6 Features

- RFC 1981 Path MTU discovery for IPv6
- RFC 2460 IPv6 specification
- RFC 2464 Transmission of IPv6 packets over Ethernet networks
- RFC 2711 IPv6 router alert option
- RFC 3484 Default address selection for IPv6
- RFC 3587 IPv6 global unicast address format
- RFC 3596 DNS extensions to support IPv6
- RFC 4007 IPv6 scoped address architecture
- RFC 4193 Unique local IPv6 unicast addresses
- RFC 4213 Transition mechanisms for IPv6 hosts and routers
- RFC 4291 IPv6 addressing architecture
- RFC 4443 Internet Control Message Protocol (ICMPv6)
- RFC 4861 Neighbor discovery for IPv6
- RFC 4862 IPv6 Stateless Address Auto-Configuration (SLAAC)
- RFC 5014 IPv6 socket API for source address selection
- RFC 5095 Deprecation of type 0 routing headers in IPv6
- RFC 5175 IPv6 Router Advertisement (RA) flags option
- RFC 6105 IPv6 Router Advertisement (RA) guard

Management

- AT Enterprise MIB including AMF Plus MIB and SNMP traps
- Optical DDM MIB
- SNMPv1, v2c and v3
- IEEE 802.1AB Link Layer Discovery Protocol (LLDP)
- RFC 1155 Structure and identification of management information for TCP/IP-based Internets
- RFC 1157 Simple Network Management Protocol (SNMP)
- RFC 1212 Concise MIB definitions
- RFC 1213 MIB for network management of TCP/IP-based Internets: MIB-II
- RFC 1215 Convention for defining traps for use with the SNMP
- RFC 1227 SNMP MUX protocol and MIB
- RFC 1239 Standard MIB
- RFC 1724 RIPv2 MIB extension
- RFC 2578 Structure of Management Information v2 (SMIv2)
- RFC 2579 Textual conventions for SMIv2
- RFC 2580 Conformance statements for SMIv2
- RFC 2674 Definitions of managed objects for bridges with traffic classes, multicast filtering and VLAN extensions
- RFC 2741 Agent extensibility (AgentX) protocol
- RFC 2787 Definitions of managed objects for VRRP
- RFC 2819 RMON MIB (groups 1,2,3 and 9)
- RFC 2863 Interfaces group MIB

- RFC 3176 sFlow: a method for monitoring traffic in switched and routed networks
- RFC 3411 An architecture for describing SNMP management frameworks
- RFC 3412 Message processing and dispatching for the SNMP
- RFC 3413 SNMP applications
- RFC 3414 User-based Security Model (USM) for SNMPv3
- RFC 3415 View-based Access Control Model (VACM) for SNMP
- RFC 3416 Version 2 of the protocol operations for the SNMP
- RFC 3417 Transport mappings for the SNMP
- RFC 3418 MIB for SNMP
- RFC 3621 Power over Ethernet (PoE) MIB
- RFC 3635 Definitions of managed objects for the Ethernet-like interface types
- RFC 3636 IEEE 802.3 MAU MIB
- RFC 4022 MIB for the Transmission Control Protocol (TCP)
- RFC 4113 MIB for the User Datagram Protocol (UDP)
- RFC 4188 Definitions of managed objects for bridges
- RFC 4292 IP forwarding table MIB
- RFC 4293 MIB for the Internet Protocol (IP)
- RFC 4318 Definitions of managed objects for bridges with RSTP
- RFC 4560 Definitions of managed objects for remote ping, traceroute and lookup operations
- RFC 5424 Syslog protocol
- RFC 6527 Definitions of managed objects for VRRPv3

Multicast Support

- Bootstrap Router (BSR) mechanism for PIM-SM
- IGMP query solicitation
- IGMP snooping (IGMPv1, v2 and v3)
- IGMP snooping fast-leave
- IGMP/MLD multicast forwarding (IGMP/MLD proxy)
- MLD snooping (MLDv1 and v2)
- PIM-SM and PIM-SSM for IPv6
- RFC 1112 Host extensions for IP multicasting (IGMPv1)
- RFC 2236 Internet Group Management Protocol v2 (IGMPv2)
- RFC 2710 Multicast Listener Discovery (MLD) for IPv6
- RFC 2715 Interoperability rules for multicast routing protocols
- RFC 3306 Unicast-prefix-based IPv6 multicast addresses
- RFC 3376 IGMPv3
- RFC 3618 Multicast Source Discovery Protocol (MSDP)
- RFC 3810 Multicast Listener Discovery v2 (MLDv2) for IPv6
- RFC 3956 Embedding the Rendezvous Point (RP) address in an IPv6 multicast address
- RFC 3973 PIM Dense Mode (DM)
- RFC 4541 IGMP and MLD snooping switches
- RFC 4601 Protocol Independent Multicast - Sparse Mode (PIM-SM): protocol specification (revised)
- RFC 4604 Using IGMPv3 and MLDv2 for source-specific multicast
- RFC 4607 Source-specific multicast for IP

Open Shortest Path First (OSPF)

- OSPF link-local signaling
- OSPF MD5 authentication
- Out-of-band LSDB resync
- RFC 1245 OSPF protocol analysis
- RFC 1246 Experience with the OSPF protocol
- RFC 1370 Applicability statement for OSPF
- RFC 1765 OSPF database overflow
- RFC 2328 OSPFv2
- RFC 2370 OSPF opaque LSA option
- RFC 2740 OSPFv3 for IPv6
- RFC 3101 OSPF Not-So-Stubby Area (NSSA) option
- RFC 3509 Alternative implementations of OSPF area border routers
- RFC 3623 Graceful OSPF restart
- RFC 3630 Traffic engineering extensions to OSPF
- RFC 4552 Authentication/confidentiality for OSPFv3
- RFC 5329 Traffic engineering extensions to OSPFv3
- RFC 5340 OSPFv3 for IPv6 (partial support)

Quality of Service (QoS)

- IEEE 802.1p Priority tagging
- RFC 2211 Specification of the controlled-load network element service

RFC 2474	DiffServ precedence for eight queues/port	IEEE 802.1X	Multi-supplicant authentication	RFC 858	Telnet suppress go ahead option
RFC 2475	DiffServ architecture	IEEE 802.1X	Port-based network access control	RFC 1091	Telnet terminal-type option
RFC 2597	DiffServ Assured Forwarding (AF)	RFC 2560	X.509 Online Certificate Status Protocol(OCSP)	RFC 1350	Trivial File Transfer Protocol (TFTP)
RFC 2697	A single-rate three-color marker	RFC 2818	HTTP over TLS ("HTTPS")	RFC 1985	SMTP service extension
RFC 2698	A two-rate three-color marker	RFC 2865	RADIUS authentication	RFC 2049	MIME
RFC 3246	DiffServ Expedited Forwarding (EF)	RFC 2866	RADIUS accounting	RFC 2131	DHCPv4 (server, relay and client)
		RFC 2868	RADIUS attributes for tunnel protocol support	RFC 2132	DHCP options and BootP vendor extensions
		RFC 2986	PKCS #10: certification request syntax specification v1.7	RFC 2616	Hypertext Transfer Protocol - HTTP/1.1
				RFC 2821	Simple Mail Transfer Protocol (SMTP)
		RFC 3546	Transport Layer Security (TLS) extensions	RFC 2822	Internet message format
		RFC 3579	RADIUS support for Extensible Authentication Protocol (EAP)	RFC 3046	DHCP relay agent information option (DHCP option 82)
		RFC 3580	IEEE 802.1x RADIUS usage guidelines	RFC 3315	DHCPv6 (server, relay and client)
		RFC 3748	PPP Extensible Authentication Protocol (EAP)	RFC 3633	IPv6 prefix options for DHCPv6
		RFC 4251	Secure Shell (SSHv2) protocol architecture	RFC 3646	DNS configuration options for DHCPv6
		RFC 4252	Secure Shell (SSHv2) authentication protocol	RFC 3993	Subscriber-ID suboption for DHCP relay agent option
		RFC 4253	Secure Shell (SSHv2) transport layer protocol		
		RFC 4254	Secure Shell (SSHv2) connection protocol	RFC 4330	Simple Network Time Protocol (SNTP) version 4
		RFC 5176	RADIUS CoA (Change of Authorization)	RFC 5905	Network Time Protocol (NTP) version 4
		RFC 5280	X.509 certificate and Certificate Revocation List (CRL) profile		
		RFC 5425	Transport Layer Security (TLS) transport mapping for Syslog		
		RFC 5656	Elliptic curve algorithm integration for SSH		
		RFC 6125	Domain-based application service identity within PKI using X.509 certificates with TLS		
		RFC 6614	Transport Layer Security (TLS) encryption for RADIUS		
		RFC 6668	SHA-2 data integrity verification for SSH		
		RFC 8446	Transport Layer Security (TLS) v1.3		
		RFC 8894	Simple Certificate Enrollment Protocol (SCEP)		

Resiliency Features

- IEC 61439-2 Media Redundancy Protocol (MRP)
- ITU-T G.8032 / Y.1344 Ethernet Ring Protection Switching (ERPS)
- IEEE 802.Tag CFM Continuity Check Protocol (CCP)
- IEEE 802.1AX Link aggregation (static and LACP)
- IEEE 802.1D MAC bridges
- IEEE 802.1s Multiple Spanning Tree Protocol (MSTP)
- IEEE 802.1w Rapid Spanning Tree Protocol (RSTP)
- IEEE 802.3ad Static and dynamic link aggregation
- RFC 5798 Virtual Router Redundancy Protocol version 3 (VRRPv3) for IPv4 and IPv6
- RFC5880 Bidirectional Forwarding Detection (BFD)

Routing Information Protocol(RIP)

- RFC 1058 Routing Information Protocol(RIP)
- RFC 2080 RIPng for IPv6
- RFC 2081 RIPng protocol applicability statement
- RFC 2082 RIP-2 MD5 authentication
- RFC 2453 RIPv2

Security Features

- SSH remote login
- SSLv2 and SSLv3
- TACACS+ Accounting, Authentication and Authorisation (AAA)
- IEEE 802.1X Authentication protocols (TLS, TTLS, PEAP and MD5)

Feature Licenses

	Description	Includes	Stack Licensing
AT-FL-x930-01	x930 Premium license	■ OSPF² (16,000 routes) ■ BGP4² (5,000 routes) ■ PIMv4-SM, DM and SSM (2,000 entries) ■ VLAN double tagging (Q-in-Q) ■ RIPng (5,000 routes) ■ OSPFv3 (8,000 routes) ■ BGP4+ (5,000 routes) ■ MLDv1 and v2 ■ PIM-SMv6/SSMv6 (1,000 entries) ■ VRF lite (64 domains) ■ RADIUS Full ■ UDLD ■ PTP Transparent Mode	■ One license per stack member
AT-SW-APM10-1YR^{3,4}	Cumulative AMF Plus Master license	■ AMF Plus Master license for up to 10 nodes for 1 year	■ One license per stack
AT-SW-APM10-5YR^{3,4}	Cumulative AMF Plus Master license	■ AMF Plus Master license for up to 10 nodes for 5 years	■ One license per stack
AT-FL-x930-OF13-1YR	OpenFlow license	■ OpenFlow v1.3 for 1 year	■ Not supported on a stack
AT-FL-x930-OF13-5YR	OpenFlow license	■ OpenFlow v1.3 for 5 years	■ Not supported on a stack
AT-FL-x930-AAP-1YR	AMF Application Proxy license	■ License that enables communication with an AMF-Security server for 1 year	■ One license per stack
AT-FL-x930-AAP-5YR	AMF Application Proxy license	■ License that enables communication with an AMF-Security server for 5 years	■ One license per stack
AT-FL-x930-8032	ITU-T G.8032 license	■ G.8032 ring protection ■ Ethernet CFM	■ One license per stack member
AT-FL-x930-CPOE	Continuous POE license	■ Continuous PoE power for GPX models only	■ One license per stack member
AT-FL-x930-MSEC⁵	MACSec license	■ Media Access Control Security	■ One license per stack member
AT-FL-x930-MODB	Modbus license	■ Modbus for industrial applications	■ One license per stack member

² The standard switch software supports 64 OSPF and BGP routes

³ From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide all standard AMF network management and automation features. They also enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards)

⁴ Purchase one license per 10 nodes (up to 120 nodes maximum)

⁵ MACSec is only supported on 1GbE downlink ports

Feature Licenses continued

	Description	Includes	Stack Licensing
AT-FL-x930-MRP	MRP license	■ Media Redundancy Protocol	■ One license per stack member
AT-SW-AWC10-1YR⁶	Cumulative AWC license	■ Autonomous Wave Control (AWC) license for up to 10 access points for 1 year	■ One license per stack
AT-SW-AWC10-5YR⁶	Cumulative AWC license	■ Autonomous Wave Control (AWC) license for up to 10 access points for 5 years	■ One license per stack
AT-SW-CB10-1YR-2022⁷	Cumulative AWC-CB and AWC-SC license	■ AWC Channel Blanket and AWC Smart Connect license for up to 10 access points for 1 year	■ One license per stack
AT-SW-CB10-5YR-2022⁷	Cumulative AWC-CB and AWC-SC license	■ AWC Channel Blanket and AWC Smart Connect license for up to 10 access points for 5 years	■ One license per stack

⁶ 5 APs can be managed for free. Purchase one license per 10 additional APs (up to 120 APs maximum)

⁷ Channel Blanket and Smart Connect are not available as free services. Both an AWC-CB license and an AWC license are required for Channel Blanket and/or Smart Connect to operate. Purchase one AWC-CB license per 10 APs (up to 180 APs maximum).

ORDERING INFORMATION

AT-x930-28GTX-BOy^{8,9,10}	24-port 10/100/1000T stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GPX-BOy^{8,9,10}	24-port 10/100/1000T PoE+ stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GSTX-BOy^{8,9,10}	24-port 10/100/1000T and 24-port 100/1000 SFP stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-52GTX-BOy^{8,9,10}	48-port 10/100/1000T stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-52GPX-BOy^{8,9,10}	48-port 10/100/1000T PoE+ stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GTX^{9,10}	24-port 10/100/1000T stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GPX^{9,10}	24-port 10/100/1000T PoE+ stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GSTX^{9,10}	24-port 10/100/1000T and 24-port 100/1000 SFP stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-52GTX^{9,10}	48-port 10/100/1000T stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-52GPX^{9,10}	48-port 10/100/1000T PoE+ stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GTX-J90*	24-port 10/100/1000T stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GPX-J90*	24-port 10/100/1000T PoE+ stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-28GSTX-J90*	24-port 10/100/1000T and 24-port 100/1000 SFP stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-52GTX-J90*	48-port 10/100/1000T stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-x930-52GPX-J90*	48-port 10/100/1000T PoE+ stackable switch with 4 SFP+ ports and dual hotswap PSU bays
AT-RKMT-SL01	Sliding rack mount kit

Where x = 10 for US power cord
20 for no power cord
30 for UK power cord
40 for AU power cord
50 for EU power cord

⁸ Where Oy = 01 for 1 year Net Cover support
05 for 5 years Net Cover support

⁹ Power supplies must be ordered separately

¹⁰ Trade Act Agreement compliant

* J1TC approved models only available in North America. Please contact your sales representative.

Accessories

Expansion Module	
AT-StackQS	2 x QSFP+ expansion module
AT-x9EM/XT4	4 x 10GBASE-T expansion module
Power Supplies (for all models)	
AT-PWR150-xx⁹	150W system power supply
AT-PWR250 v2-xx^{9, 11}	250W system power supply
AT-PWR250-80⁹	250W DC system power supply
AT-PWR800-xx^{9, 12}	800W PoE+ power supply
AT-PWR800 v2-xx^{9, 12}	800W PoE+ power supply
AT-PWR1200-xx⁹	1200W PoE+ power supply
AT-PWR1200 v2-xx⁹	1200W PoE+ power supply
Fan accessories	
AT-FAN09	Spare x930 fan module
AT-FAN09ADP	Spare x930 fan adaptor board
40G QSFP+ Modules For use with AT-StackQS module	
AT-QSFP1CU	1 meter QSFP+ direct attach stacking cable
AT-QSFPSR4	40GSR4 850 nm short-haul up to 150 m with MMF
AT-QSFPLR4	40GLR4 1310 nm medium-haul, 10 km with SMF
AT-QSFPER4	40GER4 1310 nm long-haul, 40 km with SMF
AT-QSFPSR	40GSR 850 nm short-haul up to 150 m with MMF

⁹ Power supplies must be ordered separately

¹⁰ Trade Act Agreement compliant

¹¹ Available in the APAC region only

¹² The x930 does not support mixed use of PWR800 and PWR800 v2

Breakout Cables For 4 x 10G connections	
AT-QSFP-4SFP10G-3CU	QSFP to 4 x SFP+ breakout direct attach cable (3m)
AT-QSFP-4SFP10G-5CU	QSFP to 4 x SFP+ breakout direct attach cable (5m)
10GbE SFP+ Modules (Note that any Allied Telesis 10G SFP+ module can be used for stacking with the front panel 10G ports)	
AT-SP10SR	10GSR 850 nm short-haul, 300 m with MMF
AT-SP10SR/I	10GSR 850 nm short-haul, 300 m with MMF industrial temperature
AT-SP10LRa/I	10GLR 1310 nm medium-haul, 10 km with SMF industrial temperature
AT-SP10ER40a/I	10GER 1310 nm long-haul, 40 km with SMF industrial temperature
AT-SP10BD10/I-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 10 km industrial temperature, TAA ¹⁰
AT-SP10BD10/I-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 10 km industrial temperature, TAA ¹⁰
AT-SP10BD20-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 20 km, TAA ¹⁰
AT-SP10BD20-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 20 km, TAA ¹⁰
AT-SP10BD40/I-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 40 km industrial temperature, TAA ¹⁰
AT-SP10BD40/I-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 40 km industrial temperature, TAA ¹⁰
AT-SP10TM	1G/2.5G/5G/10G, 100 m copper, TAA ¹⁰
AT-SP10TW1	1 meter SFP+ direct attach cable
AT-SP10TW3	3 meter SFP+ direct attach cable
AT-SP10TW7	7 meter SFP+ direct attach cable

1000Mbps SFP Modules	
AT-SPSX/I	1000SX GbE multi-mode 850 nm fiber up to 550 m industrial temperature, TAA ¹⁰
AT-SPTXc	1000T 100 m copper
AT-SPSX	1000SX GbE multi-mode 850 nm fiber up to 550 m
AT-SPEX	1000X GbE multi-mode 1310 nm fiber up to 2 km
AT-SPLX10a	1000LX GbE single-mode 1310 nm fiber up to 10 km
AT-SPLX10a/I	1000LX GbE single-mode 1310 nm fiber up to 10 km industrial temperature
AT-SPBD10-13	1000LX GbE Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 10 km
AT-SPBD10-14	1000LX GbE Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 10 km
AT-SPBD20-13/I	1000BX GbE Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 20 km industrial temperature
AT-SPBD20-14/I	1000BX GbE Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 20 km industrial temperature

AT-SPBD40-13/I	1000LX GbE single-mode Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 40 km, industrial temperature
AT-SPBD40-14/I	1000LX GbE single-mode Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 40 km, industrial temperature
AT-SPLX40	1000LX GbE single-mode 1310 nm fiber up to 40 km
AT-SPZX80	1000ZX GbE single-mode 1550 nm fiber up to 80 km
AT-SPZX120/I	1000ZX GbE single-mode 1550 nm fiber up to 120 km industrial temperature
100Mbps SFP Modules 100Mbps SFP modules are only compatible with the SFP ports on the AT-x930-28GSTX switch)	
AT-SPFX/2	100FX multi-mode 1310 nm fiber up to 2 km
AT-SPFX/15	100FX single-mode 1310 nm fiber up to 15 km
AT-SPFXBD-LC-13	100BX Bi-Di (1310 nm Tx, 1550 nm Rx) fiber up to 10 km
AT-SPFXBD-LC-15	100BX Bi-Di (1550 nm Tx, 1310 nm Rx) fiber up to 10 km

¹⁰ Trade Act Agreement compliant